

Beyond Convergence: An Empirical Analysis of Learned Q-Functions in Model-Free RL

Applied Reinforcement Learning - Group 6

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- Introduction
- Background
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- Experiments and Results
- Conclusion and Outlook

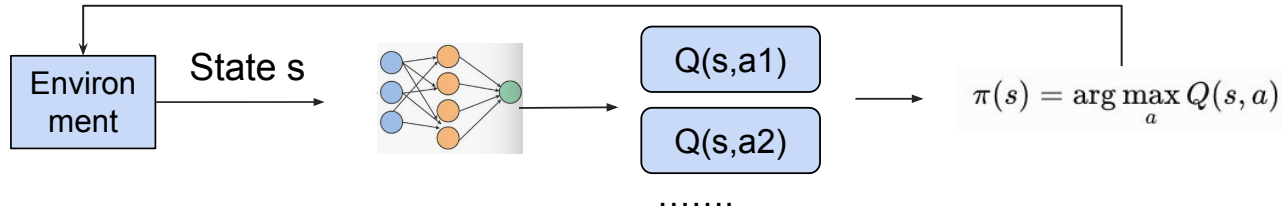
Model-free RL algorithms rely on **Q-values** to estimate future returns and improve their policies.

Yet, **evaluation usually focuses on policy performance** rather than the accuracy of these value estimates.

This work asks **whether learned Q-values accurately reflect realised returns**, and whether they become more reliable once policy performance has converged.

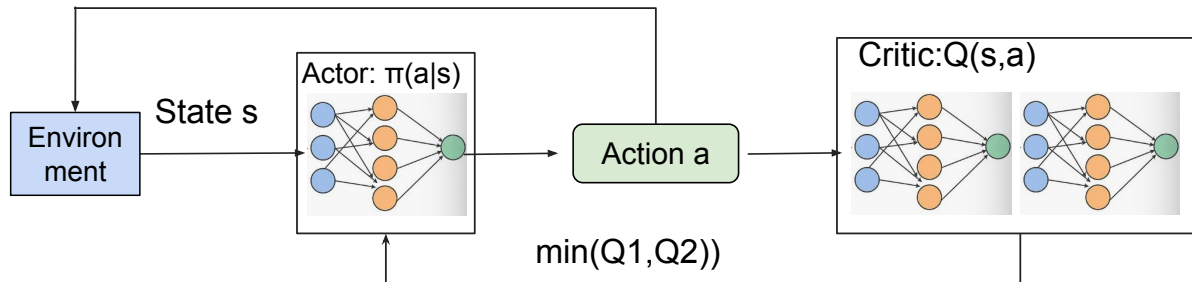
Background

Selected Algorithms



DQN

- $Q(s, a)$ via NN/MLP
- **Discrete** action space



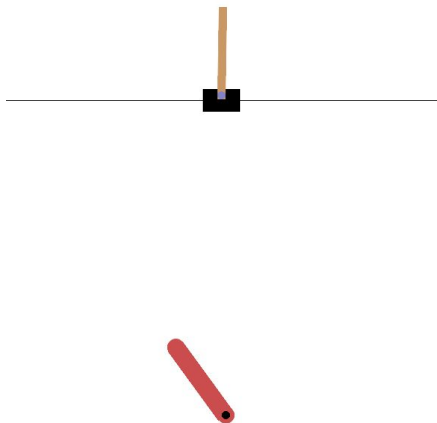
Shared: Model-free · off policy · neural networks

SAC & TD3

- Actor learns
 - stochastic policy (sac)
 - Deterministic policy (TD3)
- Twin critics learn $Q(s, a)$
- Continuous action space

Background

Benchmark Environment



	Cartpole-v1	Pendulum-v1
State Variables	$s = [x, \dot{x}, \theta, \dot{\theta}]$	$s = [\cos(\theta), \sin(\theta), \dot{\theta}]$
Action Space	$A = \{\text{push left, push right}\}$	torque $\tau \in [-2, 2]$
Reward	+1 per step	$r = -(\theta^2 + 0.1\dot{\theta}^2 + 0.001\tau^2),$
Task	Keep the pole upright	Swing into the upright position
Termination	• 500 steps • $\theta \geq 12^\circ$ • $x \geq 2.4$	200 steps without early termination

Related Work

Overestimation Bias

- Noisy value estimates combined with the max operator give systematic overestimation in Q-learning
- Solutions: Double Q-learning (decoupling), TD3 (twin critics with a min target)
- DQN also substantially overestimates on Atari games, decoupling reduces the bias
- New stabilization techniques became the new standard

Related Work

Q-Value Accuracy and Policy Performance

- Deadly Triad: function approximation, bootstrapping and off-policy learning can make value estimates diverge
- Unrealistic and growing Q-value estimates correlate with poor policy performance
- In some runs with unrealistic Q-value estimates still perform well (action rankings stay intact)
- High Q-value error $\neq$ poor policy performance

- Q-value accuracy is measured against Monte-Carlo Returns as averaged numbers
- Open questions: where in the state space errors sit, what happens after policy performance convergence
- Our approach: predicted Q-values against realized returns across algorithms and (shared) environments, tracked per checkpoint and per state

	DQN	SAC	TD3
Environment	CartPole-v1/Pendulum-v1	Pendulum-v1	Pendulum-v1

- Pendulum is discretized for DQN into $N = 11$ evenly-spaced torque bins.
- Each model is trained with the MlpPolicy and the default hyperparameters of the respective SB3 implementation.
- Checkpoint is saved every 50,000 steps

Q-value extraction:

- **DQN:** Q-value read directly from a forward pass of the Q-network for the chosen action
- **SAC/TD3 (twin-critic):** state-action pair is passed through both critic networks; report $Q1$ and $Q2$ individually as well as $\min(Q1, Q2)$

Monte Carlo return:

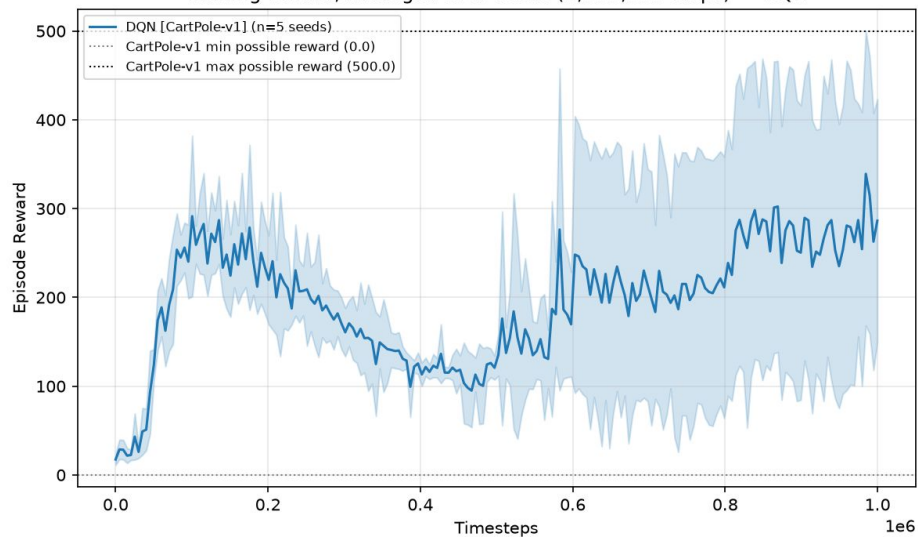
- Discounted return of one complete, deterministically executed rollout episode:

$$G(s_0) = \sum_{t=0}^T \gamma^t r_t,$$

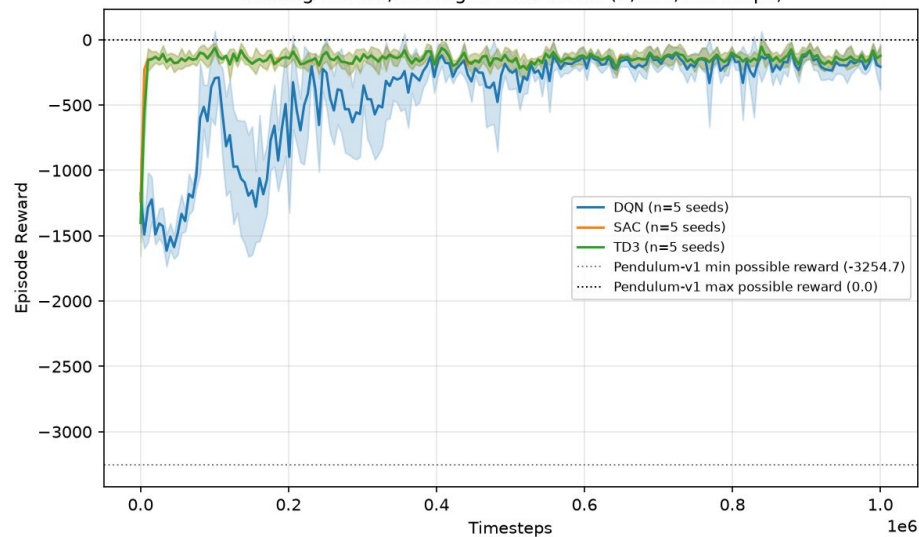
Experiments and Results

Convergence and Bias at End of Training

Training reward, averaged over seeds (1,000,000 steps) — DQN



Training reward, averaged over seeds (1,000,000 steps)



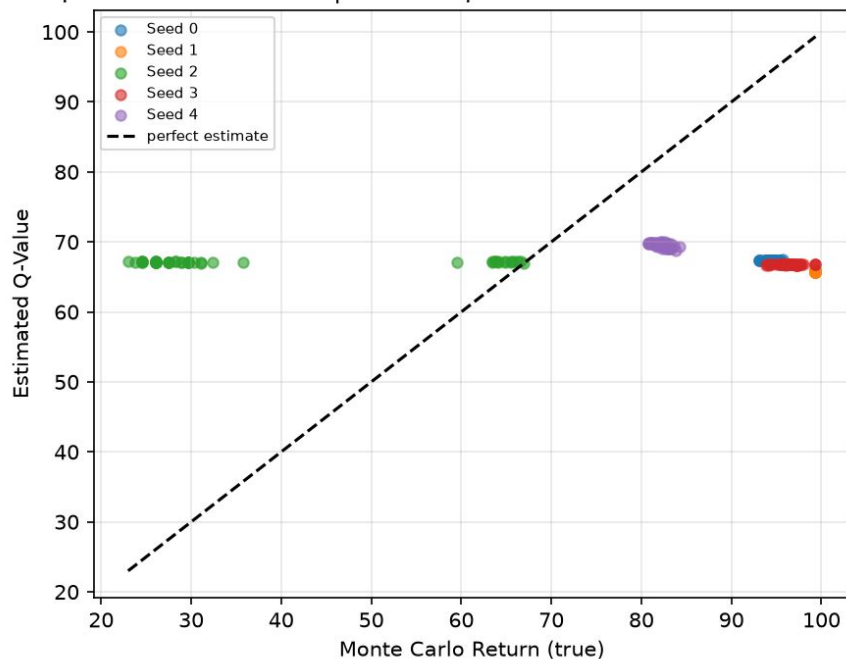
Experiments and Results

Convergence and Bias at End of Training

Q-Value Bias over 5 Seeds (1,000,000 Steps)

DQN (CartPole-v1)

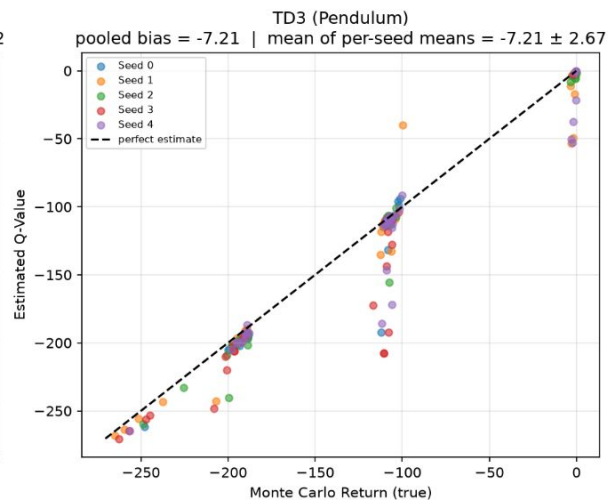
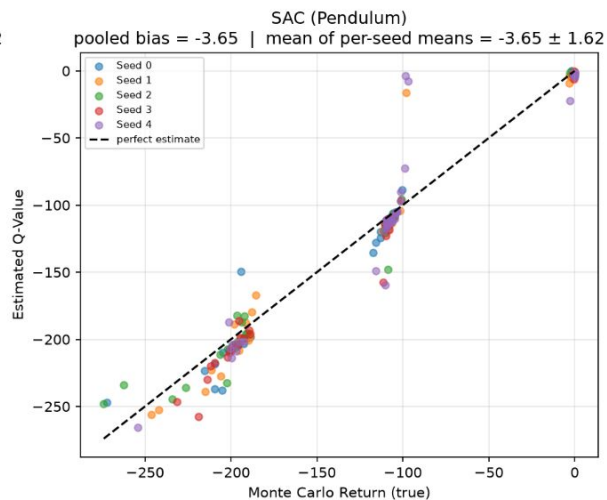
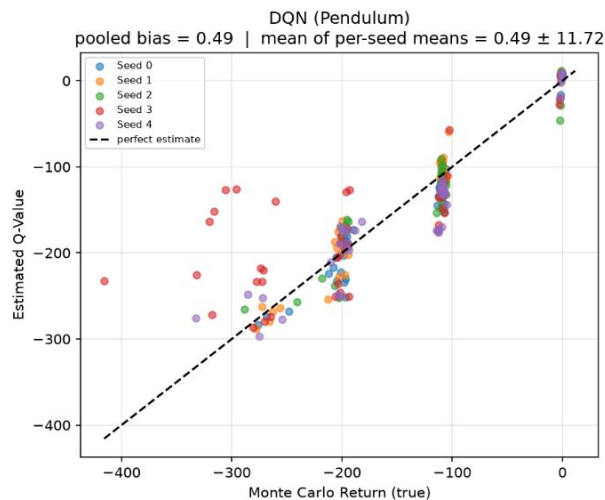
pooled bias = -14.87 | mean of per-seed means = $-14.87 \pm 22.8\%$



Experiments and Results

Convergence and Bias at End of Training

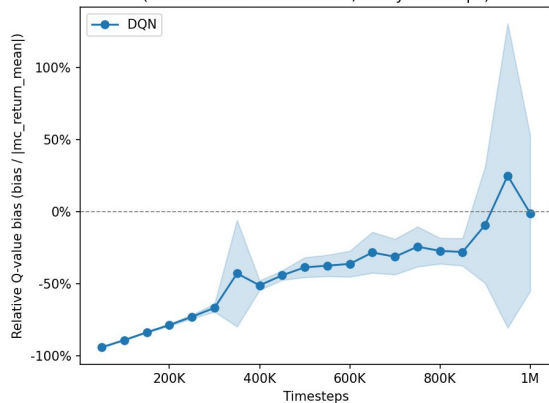
Q-Value Bias over 5 Seeds (1,000,000 Steps)



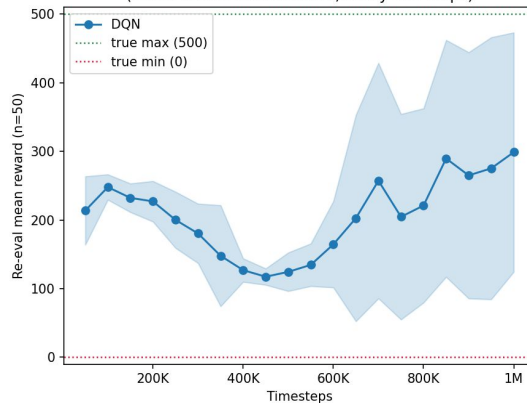
Experiments and Results

Policy Performance

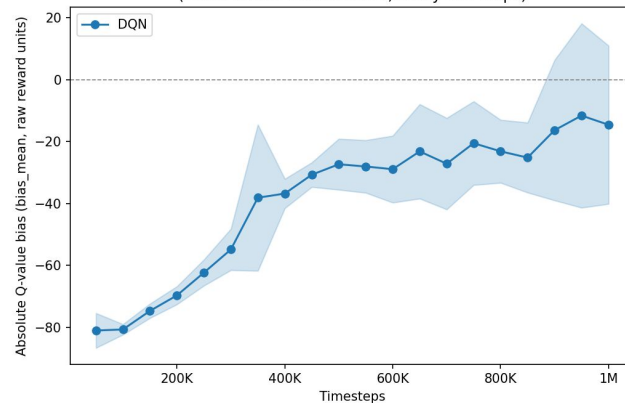
Relative Q-value bias over training
(mean ± 1 std across seeds, every 50k steps)



CartPole-v1 performance over training
(mean ± 1 std across seeds, every 50k steps)



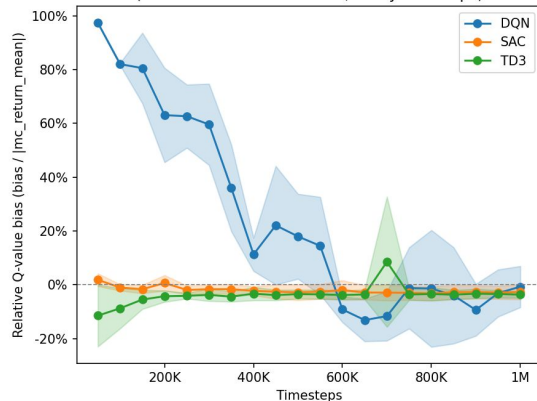
CartPole-v1: absolute Q-value bias over training
(mean ± 1 std across seeds, every 50k steps)



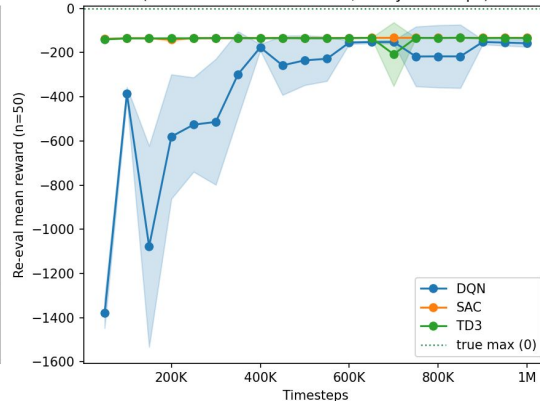
Experiments and Results

Policy Performance

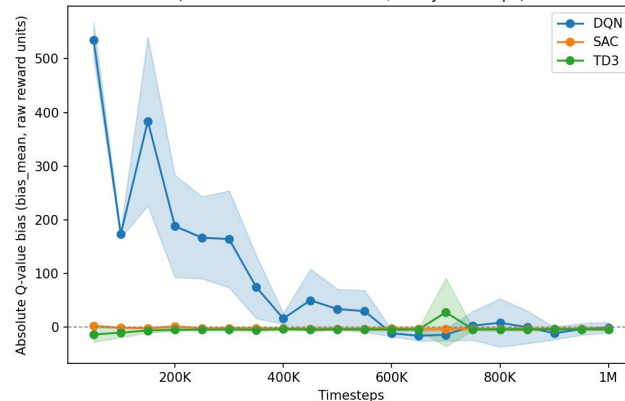
Relative Q-value bias over training
(mean ± 1 std across seeds, every 50k steps)



Pendulum-v1 performance over training
(mean ± 1 std across seeds, every 50k steps)



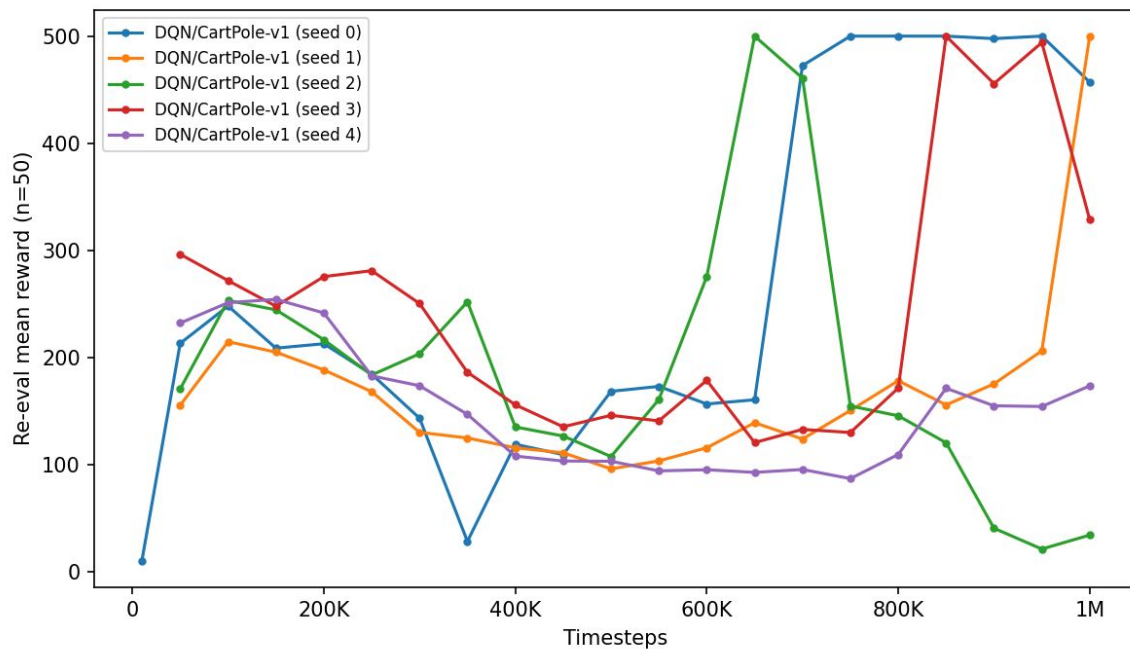
Pendulum-v1: absolute Q-value bias over training
(mean ± 1 std across seeds, every 50k steps)



Experiments and Results

Policy Performance

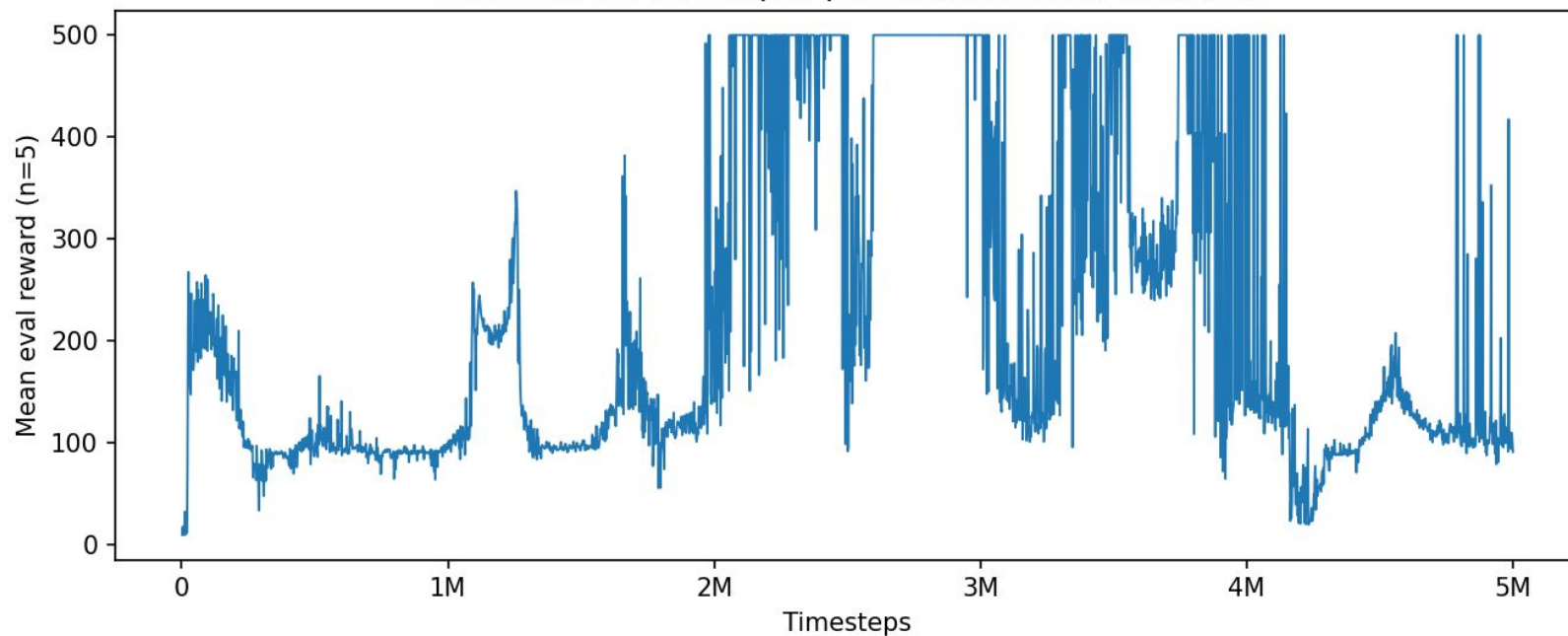
DQN/CartPole-v1: performance across seeds (n=50 re-eval)



Experiments and Results

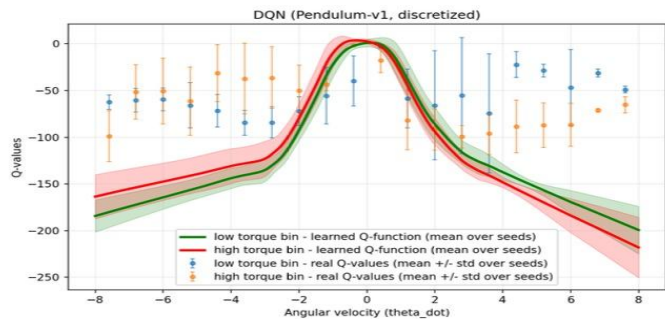
Policy Performance

DQN/CartPole (5M steps): performance never stabilizes

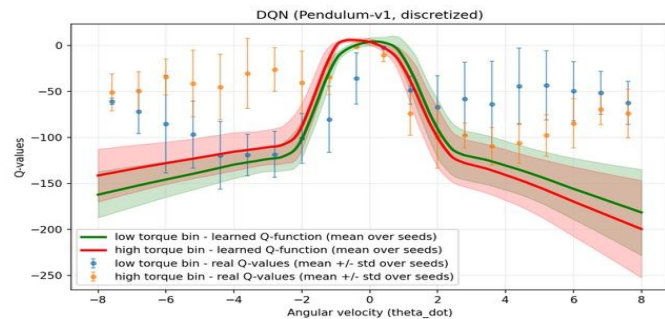


Experiments and Results

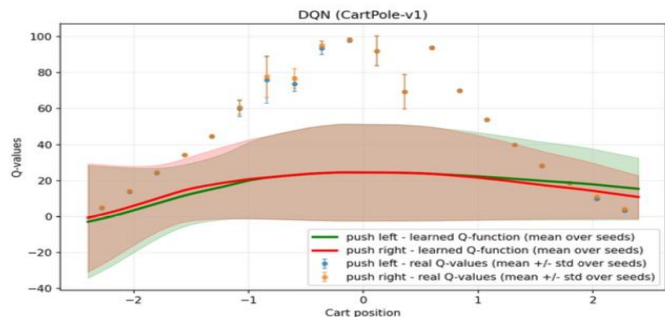
State dependence analysis (DQN/CartPole, DQN/Pendulum, SAC/Pendulum and TD3/Pendulum over 1M steps)



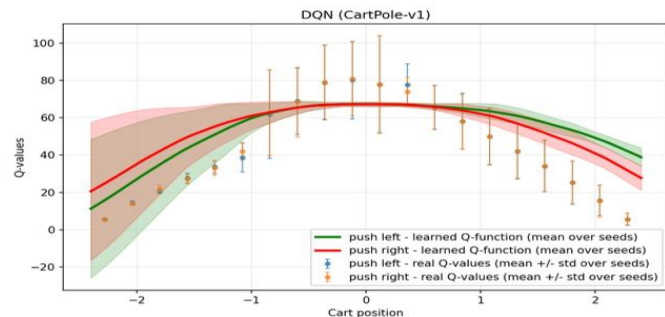
(a) Pendulum, best model



(b) Pendulum, final model



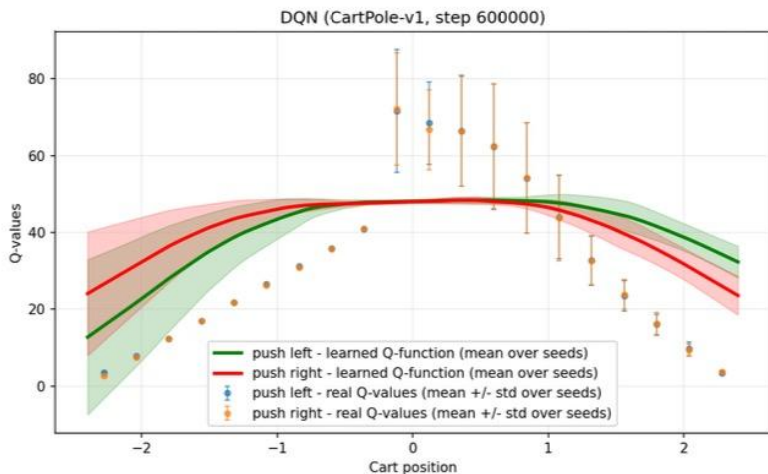
(c) CartPole, best model



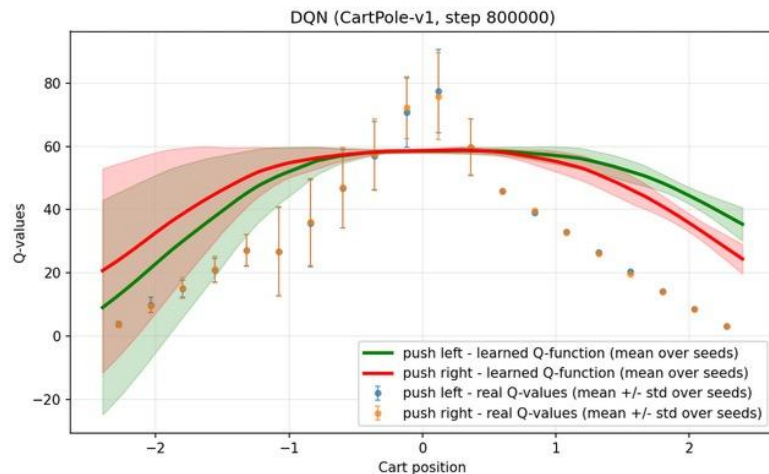
(d) CartPole, final model

Experiments and Results

State dependence analysis (DQN/CartPole over 1M steps across checkpoints)



(a) Checkpoint at 600,000

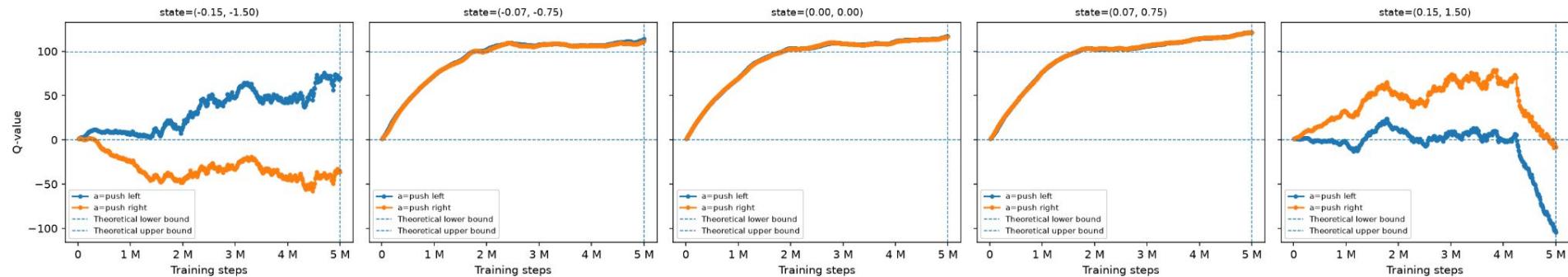


(b) Checkpoint at 800,000

Experiments and Results

Evolution of Q-functions

DQN (CartPole-v1, seed=0) -- Q-value evolution over training



Q-function stability is **state-dependent**

- Distribution of training sample shifted
- Shared NN weights indirectly influencing extreme states

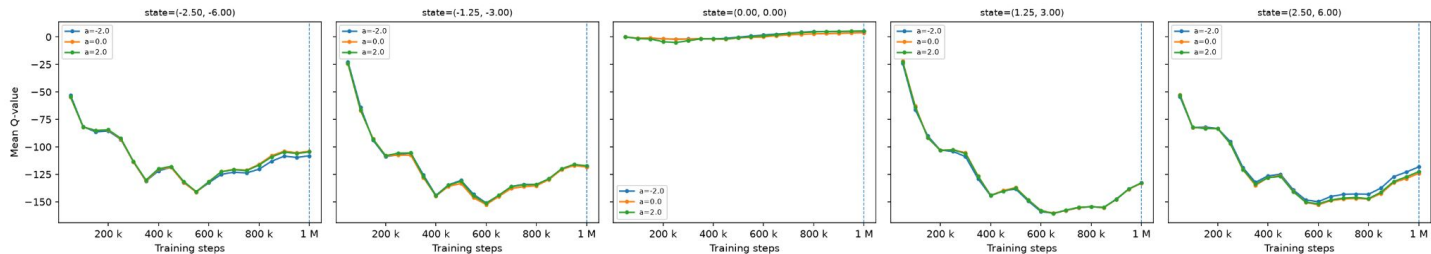
Predicted Q-function goes out of bound

$$0 \leq G \leq \sum_{t=0}^{499} \gamma^t = \frac{1 - \gamma^{500}}{1 - \gamma} \approx 99.34$$

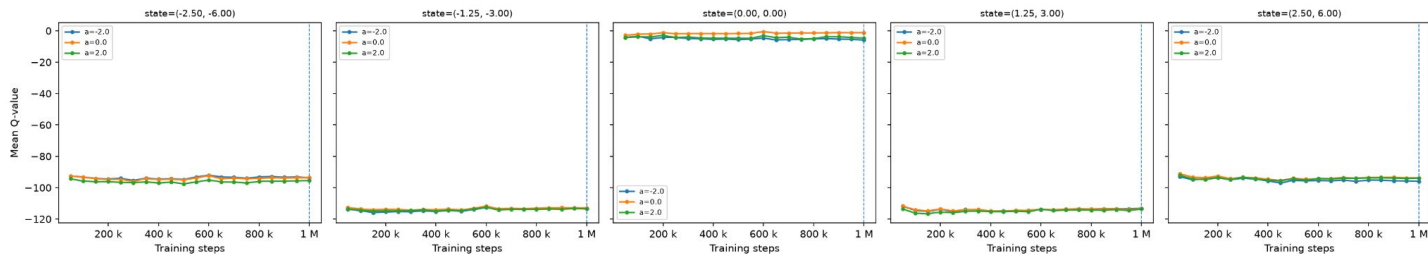
Experiments and Results

Evolution of Q-functions

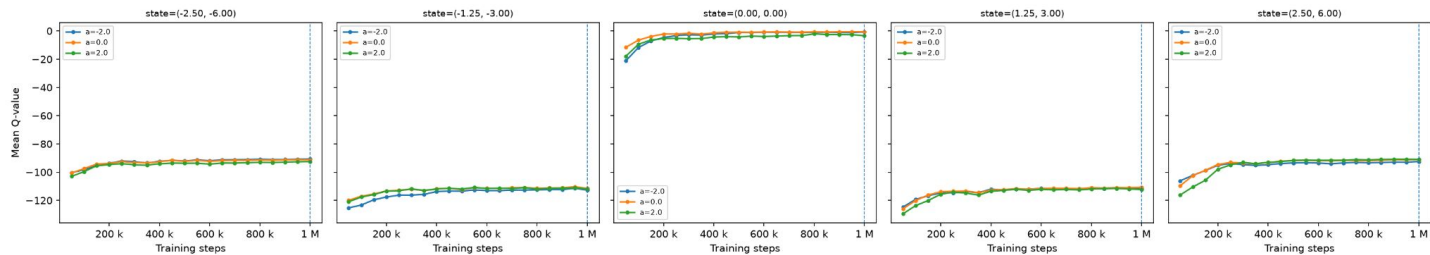
DQN (Pendulum-v1) -- Mean Q-value evolution over training (4 seeds)



SAC (Pendulum-v1) -- Mean Q-value evolution over training (5 seeds)



TD3 (Pendulum-v1) -- Mean Q-value evolution over training (5 seeds)



Conclusion and Outlook

- This work shows the learned Q-values of DQN, SAC, and TD3 on Pendulum eventually aligned well with realised returns, DQN on CartPole showed persistent value-estimation errors, which is highly state-dependent.
- **Q-function accuracy depends on the state space:**
 - Frequently visited states (central states) tend to stabilise earlier, whereas less frequently visited or more extreme states can remain unstable. A likely explanation is: as the policy improves, successful trajectories concentrate around central states, leaving extreme states less frequently sampled.
- **Longer training does not guarantee convergence**, as DQN continues to show bias and fluctuations even after millions of steps.
- Overall, evaluating **Q-values directly against empirical returns** provides valuable information that cannot be obtained from policy reward alone.

Conclusion and Outlook

- Future work could extend this analysis to **more environments, algorithms, and higher-dimensional tasks** to determine how general these observations are.
- A denser evaluation across the state space could also help clarify how Q-function accuracy relates to **state visitation and replay-buffer coverage**.
- Finally, the interaction between **Q-function accuracy, policy changes, and training dynamics** could help explain why high-performing policies can coexist with inaccurate value estimates—and potentially lead to more reliable evaluation and training methods.

Thank you for your attention!

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